

Section A: Objective Questions (22 marks)

[MCQ: +1 correct, 0 for incorrect; MSQ: +2 if all correct, -1 for incorrect]

Q1. Which theory explicitly rejects the idea of a single general intelligence factor?

- A. Spearman *
- B. Sternberg
- ☒ C. Thurstone
- D. Binet *

(1)

Q2. A participant studies a list of words. During testing, they are unable to freely recall many words, but when given a recognition test (old/new), they correctly identify most previously studied items. What is the most plausible explanation for this pattern?

- ☒ A. The information was not encoded into long-term memory
- B. The memory trace has decayed over time
- C. Retrieval cues during recall were insufficient to access stored information
- D. Recognition relies entirely on procedural memory systems

X

Q3. Which conditions are most likely to slow down visual search performance? (MSQ – Select ALL that apply)

- A. Conjunction search tasks
- ☒ B. Increased similarity between target and distractors
- ☒ C. Feature search with a unique feature
- ☒ D. High perceptual clutter
- E. Increasing the brightness or salience of the target

(-1)

(a not marked; c is incorrect)

Q4. A patient can no longer form new memories after a brain injury. However, over repeated trials, their performance on a mirror-tracing task improves, even though they insist they've never done the task before. Which explanation BEST accounts for this pattern?

- A. Episodic memory is intact but procedural memory is impaired *
- ☒ B. Procedural memory is intact but episodic memory is impaired
- C. Working memory is intact but long-term memory is impaired
- D. Semantic memory is impaired but episodic memory is intact

Declarative
↓
Episodic Semantic

(1)

Q5. Which processes are most likely to enhance long-term memory encoding? (MSQ – Select ALL that apply)

- ☒ A. Elaborative rehearsal
- B. Maintenance rehearsal alone
- ☒ C. Creating multiple retrieval cues
- D. Passive exposure without attention
- E. Learning highly similar material

(2)

→ Q6. Searching for a red object among blue objects is typically faster than searching for a red square among red circles and blue squares because:

- ☒ A. It requires top-down processing *We already model/expect what we want to find*
- ☐ B. It is a conjunction search
- ☒ C. It is a feature search (1)
- ☐ D. It involves divided attention

→ Q7. A person studies in a noisy café and later performs better on a test taken in a similar noisy environment rather than in a quiet room. Which principle is illustrated?

- ☐ A. Distributed processing
- ☒ B. Encoding specificity (1) *Best Memory fetching when required encoding matches existing encoding*
- ☐ C. Elaborative rehearsal
- ☐ D. Priming

Q8. Which statements are **TRUE** about visual search? (MSQ – Select ALL that apply)

- ☐ A. Feature search is typically faster than conjunction search
 - ☐ B. Conjunction search requires focused attention
 - ☐ C. Feature search is always serial
 - ☐ D. Increasing distractors affects conjunction search more
 - ☐ E. Conjunction search is independent of attention
- X (0) (unattempted)

Q9. Two patients are shown drawings of common objects.

Patient A is unable to copy the drawings and struggles to match objects based on shape.

Patient B can accurately copy the drawings and match shapes, but cannot identify or name the objects.

Which interpretation best explains their deficits?

- ☐ A. Patient A has associative agnosia, while Patient B has apperceptive agnosia
- ☒ B. Patient A has apperceptive agnosia, while Patient B has associative agnosia (1) *temporal lobe*
- ☐ C. Both patients have impairments in semantic memory
- ☐ D. Both patients have deficits in early visual processing

Q10. Which factors influence emotional responses to music? (MSQ – Select ALL that apply)

- ☒ A. Musical features (e.g., tempo, pitch)
- ☒ B. Listener characteristics
- ☒ C. Situational context
- ☐ D. Only lyrics determine emotional response
- ☒ E. Prior experiences and memory (2)

Q11. Maria is learning to drive a car. At first, she has to consciously think about every action.

After several months of practice, she can drive smoothly while having a conversation. According to Fitts and Posner's Model, she has progressed from:

- ☐ A. Associative Stage to Cognitive Stage
- ☒ B. Cognitive Stage to Autonomous Stage
- ☐ C. Autonomous Stage to Associative Stage
- ☐ D. Sensory-Motor Stage to Cognitive Stage

Cognitive
↓
Associative
↓
Autonomous

(1)

→ Q12. Which of the following findings are consistent with early selection theories of attention?
(MSQ – Select ALL that apply)

- A. Unattended information is filtered before semantic processing
- B. Physical characteristics of unattended stimuli can be reported
- C. The meaning of unattended stimuli strongly influences behaviour
- D. Processing limits occur at a late stage
- E. Selection is based on physical features

2 (0)
(unattempted)

Q13. In motor control, a "Closed-Loop" system is characterised by:

- A. Pre-planned movement without adjustment
- ☒ B. Use of sensory feedback
- C. Homunculus organisation
- D. Reflex responses

1

Q14. The "Binding Problem" explores how the brain:

- A. Misses unexpected stimuli
- ☒ B. Combines features into a single object
- C. Chooses attention filters
- D. Ignores one visual field

1

→ Q15. Which brain structure is primarily responsible for memory consolidation?

- A. Cerebellum
- ☒ B. Prefrontal Cortex
- C. Amygdala
- D. Hippocampus

1

Q16. Participants rate previously seen statements as more true than new ones, even if told some were false. Why?

- A. Strong episodic recall
- ☒ B. Familiarity increases perceived truth
- C. Effort reduction strategy
- D. Emotional valence dominates

1

Q17. A student repeatedly reads a list of random words aloud but cannot recall them a day later. Another student groups the words meaningfully and relates them to personal experiences and recalls them easily. Which combination BEST explains this difference?

- ☒ A. Maintenance rehearsal vs sensory encoding
- B. Elaborative rehearsal vs maintenance rehearsal
- C. Encoding specificity vs retrieval failure
- D. Chunking vs decay

X

P.T.O.

Section B: Short Answer Questions (3 × 4 = 12)

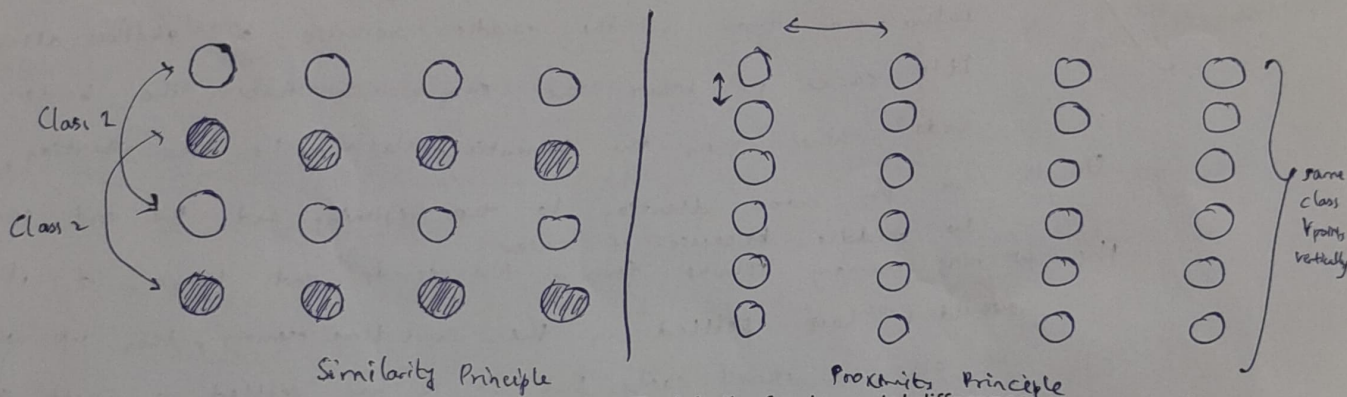
Q1. Explain any two Gestalt principles. Illustrate each with an example.

Two Gestalt Principles:

(1) Similarity → When 2 classes of objects are similar to each other, our perception is that they are of the same class.

(2) Proximity

→ When 2 ~~classes~~ ~~objects~~ of objects are closer to each other, our perception is that they belong to the same class.



Similarity Principle

Proximity Principle

Q2. Distinguish between Sensation and Perception. Explain the fundamental difference between the "raw data" gathered from the environment and the brain's "interpretation" of that data, including the concept of perception as the brain's "best guess".

At least 1 more diff

Sensation : Input of Raw Data

Perception : Understanding the Raw Data by the brain

Raw Data : This is the ~~the~~ information we can see, hear, smell, taste or feel.
e.g. Photons ~~from~~ observed from the Retina

Brain's Interpretation : This is the information when converted to Electrical signals, i.e. in the form the brain requires.

e.g. Photons from Retina → Electrical Signals

Pressure waves on Cochlea → Electrical signals

The brain combines the electrical signals received along with any previous instance of whatever is sensed, and gives its best guess.

Hence, familiarity also plays a role here. But if there is no instance of it being seen before, the brain will match it to anything which is semantically very close to the thing at hand.

Q3. Describe the "Primacy" and "Recency" effects in memory. Identify why we tend to remember words at the beginning and end of a list better than those in the middle, and explain what happens to the recency effect if there is a 30-second filled delay before recall.

Primacy Effect: When we retain something in the memory for a long period of time after learning

Recency effect: When we retain something we learnt in the memory for a short period of time, i.e. only recent events.

We tend to remember words from the beginning and end of a list rather than those in the middle because of shallow attention. It's easier to learn the easy-visible features like borders, i.e. ends rather than the semantic relation, i.e. deep attention.

Usually we pay more attention to the beginning and the end compared to middle because of ease. The working memory retains for a few seconds and hence, if it wouldn't have settled in the short-term memory, then we won't be able to recall easily. But if it has settled in short term memory, then we will be able to show recency effect.

But as it is filled delay, it affects recency effect since attention is shifted as distractions are presented.

Section C: Long Answer Questions ($2 \times 8 = 16$)

Q1. A participant studies a list of related words (e.g., bed, rest, dream, tired) but later confidently recalls the word "sleep," which was not presented. Even when told that some memories may be incorrect, the participant remains confident.

Using this scenario, explain:

- The Deese-Roediger-McDermott (DRM) paradigm and how it leads to false recall
- The role of schema-driven reconstruction
- The distinction between familiarity and source memory
- Why are such errors difficult to inhibit
- One real-world implication (e.g., misinformation, eyewitness testimony)

(a) The words (bed, rest, dream, tired) are of similar category of {sleep}, so the participant thinks that the word sleep also was also presented.

Hence, upon recall semantic closeness is also taken into consideration as the words have not consolidated to long term memory.

(b) The participant may try to create a schema ~~out~~ in the head
i.e. going to bed → taking a rest

Hence, this might get correlated to the term 'sleep' which
makes the participant ~~confident~~ about seeing 'sleep'.

(c) Familiarity is when things the participant is familiar with and
seen/know them before have been presented to them. And
due to distributed processing, the participant thinks he
has seen 'sleep' also before. This is done by Pre-frontal Cortex.

but no context
The labels have not yet moved to ~~see~~ memory yet, Hence it
is difficult for recognition. X

(d) The semantic representation/encoding of 'sleep' is very close to
that of 'bed', since it is always known that bed is to
sleep. And this is enforced by the word 'rest'.

Sleeping on 'bed' is for 'rest', hence it is ~~is~~ unusual
that this error is difficult to reproduce?

(e) eye-witness Testimony:

This is not accurate because after the ~~so~~ incident, the witness
might still be in shock and not completely settled.

So, small clever questions framed in a questionable way
may make the participant testify things that ~~have~~ ~~was~~
haven't happened

If the questions ask for similar class of answers, then the
witness might ~~can~~ get confused and testify what ~~an~~ actually
didn't occur

Misinformation:

When a very similarly phrased argument is brought into the picture,
because of it being familiar to something similar we know,
we ~~are~~ unintentionally agree to it.

Q2. Selective attention allows us to process relevant information while ignoring distractions, but there is debate about when selection occurs during processing.

- Explain the key assumptions of early and late selection theories of attention
- Compare and contrast the two approaches
- Briefly explain how intermediate models (e.g., attenuation theory) attempt to resolve this debate
- How does the system decide on early or late selection?

(a) Broadbent states that the selection occurs in the early stages during processing before going to the memory.

Denck and Norman state that selection occurs after the late stage during processing and then filter out before going to the memory.

Treisman states that it's not about early or late, the attention for distractions is minimized/reduced, not completely 0. i.e. Attenuation. Unattended information is filtered before semantic processing.

(b) Early Selection Theory states that attention shifts to relevant information as soon as it is heard, e.g. after calling name, respond and focus on that. Processing limits occur at a later stage.

Late Selection Theory states that attention should be shifted after input is given, i.e. Take all information in and then filter out and ~~give~~ focus on relevant information.

e.g. After calling a name, listen everything after and then select.

(c) As Attention cannot be completely 0, since we are usually aware of this, Instead of just 0 or 1 (Yes/No), We give a range of magnitude. We do focus on irrelevant things as well, but not as much as on the relevant ones.

(d) The system decides on early or late selection based on memory and past experience. If the relevant information is already something familiar to us, then Early selection occurs. And if ~~the~~ relevant information ends with our name, then we ~~apply~~ late selection subconsciously.

also depends on task demands, cog. load.

7